LICHTENBAUM–SCHLESSINGER COMPLEX COMPUTES THE COTANGENT COMPLEX THROUGH DEGREE MINUS TWO

THE STACKS PROJECT AUTHORS

Editorial note. The source mixes homological H_2 notation with cohomological H^{-2}; the complex is placed in degrees -2,-1,0 as stated in2392–2417.

Editorial note. Local resolution and comparison results are retained supporting work and are not counted as additional tasks.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Simplicial polynomial resolutions, simplicial identities, ordinary Tor exact sequences and tensor-product spectral sequences, the cotangent transitivity framework. The explicit two-skeleton, naive comparison and degree-two local calculation are retained as authored support rather than granted free.

Source reference. Original proposition 08PX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11040–11040. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 08N8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10981–10981. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 08PQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11033–11033. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 08PH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11025–11025. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 08PD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11021–11021. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 08PU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11037–11037. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two

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concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 08QA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11053–11053. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original definition 07BN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9348–9348. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original equation 08R9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11087–11087. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original equation 08QS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11070–11070. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 08QH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11060–11060. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 08Q1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11044–11044. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 08RA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11088–11088. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 07BP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9349–9349. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00S2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1024–1024. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 08R7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11085–11085. Stable Stacks tag for an expressly

allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 08PS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11035–11035. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 08RG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11094–11094. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 0194: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1639–1639. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 08NS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11000–11000. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 08P2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11010–11010. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 08P1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11009–11009. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 08Q9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11052–11052. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 08PT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11036–11036. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Lemma 0.1. *In the situation above denote L the complex (1.0.1). There is a canonical map $L_{B/A} \to L$ in $D(B)$ which induces an isomorphism $\tau_{\geq -2} L_{B/A} \to L$ in $D(B)$.*

Remark 0.2 (Resolutions). Let $A \to B$ be any ring map. Let us call an augmented simplicial A-algebra $\epsilon : P_\bullet \to B$ a *resolution of B over A* if each P_n is a polynomial

algebra and ϵ is a trivial Kan fibration of simplicial sets. If $P_\bullet \to B$ is an augmentation of a simplicial A-algebra with each P_n a polynomial algebra surjecting onto B, then the following are equivalent

(1) $\epsilon : P_\bullet \to B$ is a resolution of B over A,
(2) $\epsilon : P_\bullet \to B$ is a quasi-isomorphism on associated complexes,
(3) $\epsilon : P_\bullet \to B$ induces a homotopy equivalence of simplicial sets.

To see this use Simplicial, Lemmas 08NS, 08P2, and 08P1. A resolution $P_\bullet$ of B over A gives a cosimplicial object $U_\bullet$ of $\mathcal{C}_{B/A}$ as in Cohomology on Sites, Lemma 08Q9 and it follows that

$$L\pi_!\mathcal{F} = \mathcal{F}(P_\bullet)$$

functorially in $\mathcal{F}$, see Lemma 08PS. The (formal part of the) proof of Proposition 08PX shows that resolutions exist. We also have seen in the first proof of Lemma 08PT that the standard resolution of B over A is a resolution (so that this terminology doesn't lead to a conflict). However, the argument in the proof of Proposition 08PX shows the existence of resolutions without appealing to the simplicial computations in Simplicial, Section 08N8. Moreover, for *any* choice of resolution we have a canonical isomorphism

$$L_{B/A} = \Omega_{P_\bullet/A} \otimes_{P_\bullet,\epsilon} B$$

in $D(B)$ by Lemma 08PU. The freedom to choose an arbitrary resolution can be quite useful.

1. Comparison with Lichtenbaum-Schlessinger

Let $A \to B$ be a ring map. In [1] there is a fairly explicit determination of $\tau_{\geq -2}L_{B/A}$ which is often used in calculations of versal deformation spaces of singularities. The construction follows. Choose a polynomial algebra P over A and a surjection $P \to B$ with kernel I. Choose generators f_t, $t \in T$ for I which induces a surjection $F = \bigoplus_{t\in T} P \to I$ with F a free P-module. Let $Rel \subset F$ be the kernel of $F \to I$, in other words Rel is the set of relations among the f_t. Let $TrivRel \subset Rel$ be the submodule of trivial relations, i.e., the submodule of Rel generated by the elements $(\ldots, f_{t'}, 0, \ldots, 0, -f_t, 0, \ldots)$. Consider the complex of B-modules

$$Rel/TrivRel \longrightarrow F \otimes_P B \longrightarrow \Omega_{P/A} \otimes_P B \tag{1.0.1}$$

where the last term is placed in degree 0. The first map is the obvious one and the second map sends the basis element corresponding to $t \in T$ to $\mathrm{d}f_t \otimes 1$.

Definition 1.1. Let $A \to B$ be a ring map. Let M be a (B, B)-bimodule over A. An A-*biderivation* is an A-linear map $\lambda : B \to M$ such that $\lambda(xy) = x\lambda(y) + \lambda(x)y$.

Example 1.2. Let $A \to B$ be a ring map. In this example we will construct an "explicit" resolution $P_\bullet$ of B over A of length 2. To do this we follow the procedure of the proof of Proposition 08PX, see also the discussion in Remark 0.2.

We choose a surjection $P_0 = A[u_i] \to B$ where u_i is a set of variables. Choose generators $f_t \in P_0$, $t \in T$ of the ideal $\mathrm{Ker}(P_0 \to B)$. We choose $P_1 = A[u_i, x_t]$ with face maps d_0 and d_1 the unique A-algebra maps with $d_j(u_i) = u_i$ and $d_0(x_t) = 0$ and $d_1(x_t) = f_t$. The map $s_0 : P_0 \to P_1$ is the unique A-algebra map with $s_0(u_i) = u_i$. It is clear that

$$P_1 \xrightarrow{d_0 - d_1} P_0 \to B \to 0$$

is exact, in particular the map $(d_0, d_1) : P_1 \to P_0 \times_B P_0$ is surjective. Thus, if $P_\bullet$ denotes the 1-truncated simplicial A-algebra given by P_0, P_1, d_0, d_1, and s_0, then the augmentation $\mathrm{cosk}_1(P_\bullet) \to B$ is a trivial Kan fibration. The next step of the

procedure in the proof of Proposition 08PX is to choose a polynomial algebra P_2 and a surjection

$$P_2 \longrightarrow \mathrm{cosk}_1(P_\bullet)_2$$

Recall that

$$\mathrm{cosk}_1(P_\bullet)_2 = \{(g_0, g_1, g_2) \in P_1^3 \mid d_0(g_0) = d_0(g_1), d_1(g_0) = d_0(g_2), d_1(g_1) = d_1(g_2)\}$$

Thinking of $g_i \in P_1$ as a polynomial in x_t the conditions are

$$g_0(0) = g_1(0), \quad g_0(f_t) = g_2(0), \quad g_1(f_t) = g_2(f_t)$$

Thus $\mathrm{cosk}_1(P_\bullet)_2$ contains the elements $y_t = (x_t, x_t, f_t)$ and $z_t = (0, x_t, x_t)$. Every element G in $\mathrm{cosk}_1(P_\bullet)_2$ is of the form $G = H + (0, 0, g)$ where H is in the image of $A[u_i, y_t, z_t] \to \mathrm{cosk}_1(P_\bullet)_2$. Here $g \in P_1$ is a polynomial with vanishing constant term such that $g(f_t) = 0$ in P_0. Observe that

(1) $g = x_t x_{t'} - f_t x_{t'}$ and
(2) $g = \sum r_t x_t$ with $r_t \in P_0$ if $\sum r_t f_t = 0$ in P_0

are elements of P_1 of the desired form. Let

$$Rel = \mathrm{Ker}(\bigoplus\nolimits_{t \in T} P_0 \longrightarrow P_0), \quad (r_t) \longmapsto \sum r_t f_t$$

We set $P_2 = A[u_i, y_t, z_t, v_r, w_{t,t'}]$ where $r = (r_t) \in Rel$, with map

$$P_2 \longrightarrow \mathrm{cosk}_1(P_\bullet)_2$$

given by $y_t \mapsto (x_t, x_t, f_t)$, $z_t \mapsto (0, x_t, x_t)$, $v_r \mapsto (0, 0, \sum r_t x_t)$, and $w_{t,t'} \mapsto (0, 0, x_t x_{t'} - f_t x_{t'})$. A calculation (omitted) shows that this map is surjective. Our choice of the map displayed above determines the maps $d_0, d_1, d_2 : P_2 \to P_1$. Finally, the procedure in the proof of Proposition 08PX tells us to choose the maps $s_0, s_1 : P_1 \to P_2$ lifting the two maps $P_1 \to \mathrm{cosk}_1(P_\bullet)_2$. It is clear that we can take s_i to be the unique A-algebra maps determined by $s_0(x_t) = y_t$ and $s_1(x_t) = z_t$.

2. Functoriality

In this section we consider a commutative square

$$\begin{array}{ccc} B & \longrightarrow & B' \\ \uparrow & & \uparrow \\ A & \longrightarrow & A' \end{array} \tag{2.0.1}$$

of ring maps. We claim there is a canonical B-linear map of complexes

$$L_{B/A} \longrightarrow L_{B'/A'}$$

associated to this diagram. Namely, if $P_\bullet \to B$ is the standard resolution of B over A and $P'_\bullet \to B'$ is the standard resolution of B' over A', then there is a canonical map $P_\bullet \to P'_\bullet$ of simplicial A-algebras compatible with the augmentations $P_\bullet \to B$ and $P'_\bullet \to B'$. This can be seen in terms of the construction of standard resolutions in Simplicial, Section 08N8 but in the special case at hand it probably suffices to say simply that the maps

$$P_0 = A[B] \longrightarrow A'[B'] = P'_0, \quad P_1 = A[A[B]] \longrightarrow A'[A'[B']] = P'_1,$$

and so on are given by the given maps $A \to A'$ and $B \to B'$. The desired map $L_{B/A} \to L_{B'/A'}$ then comes from the associated maps $\Omega_{P_n/A} \to \Omega_{P'_n/A'}$.

Another description of the functoriality map can be given as follows. Let $\mathcal{C} = \mathcal{C}_{B/A}$ and $\mathcal{C}' = \mathcal{C}'_{B'/A}$ be the categories considered in Section 08PQ. There is a functor

$$u : \mathcal{C} \longrightarrow \mathcal{C}', \quad (P, \alpha) \longmapsto (P \otimes_A A', c \circ (\alpha \otimes 1))$$

where $c : B \otimes_A A' \to B'$ is the obvious map. As discussed in Cohomology on Sites, Example 08PH we obtain a morphism of topoi $g : Sh(\mathcal{C}) \to Sh(\mathcal{C}')$ and a commutative diagram of maps of ringed topoi

$$\begin{array}{ccccc} (Sh(\mathcal{C}'), \underline{B}) & \xleftarrow[h]{} & (Sh(\mathcal{C}'), \underline{B'}) & \xleftarrow[g]{} & (Sh(\mathcal{C}), \underline{B'}) \\ \downarrow{\scriptstyle \pi'} & & \downarrow{\scriptstyle \pi'} & & \downarrow{\scriptstyle \pi} \\ (Sh(*), B) & \xleftarrow{f} & (Sh(*), B') & \longleftarrow & (Sh(*), B') \end{array} \tag{2.0.2}$$

Here h is the identity on underlying topoi and given by the ring map $B \to B'$ on sheaves of rings. By Cohomology on Sites, Remark 08PD given $\mathcal{F}$ on $\mathcal{C}$ and $\mathcal{F}'$ on $\mathcal{C}'$ and a transformation $t : \mathcal{F} \to g^{-1}\mathcal{F}'$ we obtain a canonical map $L\pi_!(\mathcal{F}) \to L\pi'_!(\mathcal{F}')$. If we apply this to the sheaves

$$\mathcal{F} : (P, \alpha) \mapsto \Omega_{P/A} \otimes_P B, \quad \mathcal{F}' : (P', \alpha') \mapsto \Omega_{P'/A'} \otimes_{P'} B',$$

and the transformation t given by the canonical maps

$$\Omega_{P/A} \otimes_P B \longrightarrow \Omega_{P \otimes_A A'/A'} \otimes_{P \otimes_A A'} B'$$

to get a canonical map

$$L\pi_!(\Omega_{\mathcal{O}/A} \otimes_{\mathcal{O}} \underline{B}) \longrightarrow L\pi'_!(\Omega_{\mathcal{O}'/A'} \otimes_{\mathcal{O}'} \underline{B'})$$

By Lemma 08PU this gives $L_{B/A} \to L_{B'/A'}$. We omit the verification that this map agrees with the map defined above in terms of simplicial resolutions.

Lemma 2.1. *Assume (2.0.1) induces a quasi-isomorphism $B \otimes^{\mathbf{L}}_A A' = B'$. Then, with notation as in (2.0.2) and $\mathcal{F}' \in Ab(\mathcal{C}')$, we have $L\pi_!(g^{-1}\mathcal{F}') = L\pi'_!(\mathcal{F}')$.*

Proof. We use the results of Remark 0.2 without further mention. We will apply Cohomology on Sites, Lemma 08QA. Let $P_\bullet \to B$ be a resolution. If we can show that $u(P_\bullet) = P_\bullet \otimes_A A' \to B'$ is a quasi-isomorphism, then we are done. The complex of A-modules $s(P_\bullet)$ associated to $P_\bullet$ (viewed as a simplicial A-module) is a free A-module resolution of B. Namely, P_n is a free A-module and $s(P_\bullet) \to B$ is a quasi-isomorphism. Thus $B \otimes^{\mathbf{L}}_A A'$ is computed by $s(P_\bullet) \otimes_A A' = s(P_\bullet \otimes_A A')$. Therefore the assumption of the lemma signifies that $\epsilon' : P_\bullet \otimes_A A' \to B'$ is a quasi-isomorphism. □

Lemma 2.2. *Let $A \to B$ be a ring map. Then $\tau_{\geq -1} L_{B/A}$ is canonically quasi-isomorphic to the naive cotangent complex.*

Proof. Consider $P = A[B] \to B$ with kernel I. The naive cotangent complex $NL_{B/A}$ of B over A is the complex $I/I^2 \to \Omega_{P/A} \otimes_P B$, see Algebra, Definition 07BN. Observe that in (08R9) we have already constructed a canonical map

$$c : NL_{B/A} \longrightarrow \tau_{\geq -1} L_{B/A}$$

Consider the distinguished triangle (08QS)

$$L_{P/A} \otimes^{\mathbf{L}}_P B \to L_{B/A} \to L_{B/P} \to (L_{P/A} \otimes^{\mathbf{L}}_P B)[1]$$

associated to the ring maps $A \to A[B] \to B$. We know that $L_{P/A} = \Omega_{P/A}[0] = NL_{P/A}$ in $D(P)$ (Lemma 08QH and Algebra, Lemma 08Q1) and that $\tau_{\geq -1} L_{B/P} = I/I^2[1] = NL_{B/P}$ in $D(B)$ (Lemma 08RA and Algebra, Lemma 07BP). To show c is a quasi-isomorphism it suffices by Algebra, Lemma 00S2 and the long exact cohomology sequence associated to the distinguished triangle to show that the maps $L_{P/A} \to L_{B/A} \to L_{B/P}$ are compatible on cohomology groups with the corresponding maps $NL_{P/A} \to NL_{B/A} \to NL_{B/P}$ of the naive cotangent complex. We omit the verification. □

Remark 2.3. We can make the comparison map of Lemma 2.2 explicit in the following way. Let $P_\bullet$ be the standard resolution of B over A. Let $I = \mathrm{Ker}(A[B] \to B)$. Recall that $P_0 = A[B]$. The map of the lemma is given by the commutative diagram

$$\begin{array}{ccccccccc} L_{B/A} & & \ldots & \longrightarrow & \Omega_{P_2/A} \otimes_{P_2} B & \longrightarrow & \Omega_{P_1/A} \otimes_{P_1} B & \longrightarrow & \Omega_{P_0/A} \otimes_{P_0} B \\ \downarrow & & & & \downarrow & & \downarrow & & \downarrow \\ NL_{B/A} & & \ldots & \longrightarrow & 0 & \longrightarrow & I/I^2 & \longrightarrow & \Omega_{P_0/A} \otimes_{P_0} B \end{array}$$

We construct the downward arrow with target I/I^2 by sending $\mathrm{d}f \otimes b$ to the class of $(d_0(f) - d_1(f))b$ in I/I^2. Here $d_i : P_1 \to P_0$, $i = 0, 1$ are the two face maps of the simplicial structure. This makes sense as $d_0 - d_1$ maps P_1 into $I = \mathrm{Ker}(P_0 \to B)$. We omit the verification that this rule is well defined. Our map is compatible with the differential $\Omega_{P_1/A} \otimes_{P_1} B \to \Omega_{P_0/A} \otimes_{P_0} B$ as this differential maps $\mathrm{d}f \otimes b$ to $\mathrm{d}(d_0(f) - d_1(f)) \otimes b$. Moreover, the differential $\Omega_{P_2/A} \otimes_{P_2} B \to \Omega_{P_1/A} \otimes_{P_1} B$ maps $\mathrm{d}f \otimes b$ to $\mathrm{d}(d_0(f) - d_1(f) + d_2(f)) \otimes b$ which are annihilated by our downward arrow. Hence a map of complexes. We omit the verification that this is the same as the map of Lemma 2.2.

Remark 2.4. Adopt notation as in Remark 08R7. The arguments given there show that the differential

$$H_2(\mathcal{C}_{B/A}, \mathcal{J}/\mathcal{J}^2) \longrightarrow H_0(\mathcal{C}_{B/A}, \mathrm{Tor}_1^{\mathcal{O}}(\mathcal{J}, \underline{B}))$$

of the spectral sequence is an isomorphism. Let $\mathcal{C}'_{B/A}$ denote the full subcategory of $\mathcal{C}_{B/A}$ consisting of surjective maps $P \to B$. The agreement of the cotangent complex with the naive cotangent complex (Lemma 2.2) shows that we have an exact sequence of sheaves

$$0 \to \underline{H_1(L_{B/A})} \to \mathcal{J}/\mathcal{J}^2 \xrightarrow{\mathrm{d}} \Omega \to \underline{H_2(L_{B/A})} \to 0$$

on $\mathcal{C}'_{B/A}$. It follows that $\mathrm{Ker}(d)$ and $\mathrm{Coker}(d)$ on the whole category $\mathcal{C}_{B/A}$ have vanishing higher homology groups, since these are computed by the homology groups of constant simplicial abelian groups by Lemma 08PS. Hence we conclude that

$$H_n(\mathcal{C}_{B/A}, \mathcal{J}/\mathcal{J}^2) \to H_n(L_{B/A})$$

is an isomorphism for all $n \geq 2$. Combined with the remark above we obtain the formula $H_2(L_{B/A}) = H_0(\mathcal{C}_{B/A}, \mathrm{Tor}_1^{\mathcal{O}}(\mathcal{J}, \underline{B}))$.

Remark 2.5. Let $A \to B$ be a ring map. Let $P_\bullet$ be a resolution of B over A (Remark 0.2). Set $J_n = \mathrm{Ker}(P_n \to B)$. Note that

$$\mathrm{Tor}_2^{P_n}(B, B) = \mathrm{Tor}_1^{P_n}(J_n, B) = \mathrm{Ker}(J_n \otimes_{P_n} J_n \to J_n^2).$$

Hence $H_2(L_{B/A})$ is canonically equal to

$$\mathrm{Coker}(\mathrm{Tor}_2^{P_1}(B, B) \to \mathrm{Tor}_2^{P_0}(B, B))$$

by Remark 2.4. To make this more explicit we choose P_2, P_1, P_0 as in Example 1.2. We claim that

$$\mathrm{Tor}_2^{P_1}(B, B) = \wedge^2(\bigoplus\nolimits_{t \in T} B) \ \oplus \ \bigoplus\nolimits_{t \in T} J_0 \ \oplus \ \mathrm{Tor}_2^{P_0}(B, B)$$

Namely, the basis elements $x_t \wedge x_{t'}$ of the first summand corresponds to the element $x_t \otimes x_{t'} - x_{t'} \otimes x_t$ of $J_1 \otimes_{P_1} J_1$. For $f \in J_0$ the element $x_t \otimes f$ of the second summand corresponds to the element $x_t \otimes s_0(f) - s_0(f) \otimes x_t$ of $J_1 \otimes_{P_1} J_1$. Finally, the map $\mathrm{Tor}_2^{P_0}(B, B) \to \mathrm{Tor}_2^{P_1}(B, B)$ is given by s_0. The map $d_0 - d_1 : \mathrm{Tor}_2^{P_1}(B, B) \to$

$\mathrm{Tor}_2^{P_0}(B, B)$ is zero on the last summand, maps $x_t \otimes f$ to $f \otimes f_t - f_t \otimes f$, and maps $x_t \wedge x_{t'}$ to $f_t \otimes f_{t'} - f_{t'} \otimes f_t$. All in all we conclude that there is an exact sequence

$$\wedge_B^2(J_0/J_0^2) \to \mathrm{Tor}_2^{P_0}(B, B) \to H^{-2}(L_{B/A}) \to 0$$

In this way we obtain a direct proof of a consequence of Quillen's spectral sequence discussed in Remark 08RG.

Lemma 2.6. *Let $P = A[S]$ be a polynomial ring over A. Let M be a (P, P)-bimodule over A. Given $m_s \in M$ for $s \in S$, there exists a unique A-biderivation $\lambda : P \to M$ mapping s to m_s for $s \in S$.*

Proof. We set

$$\lambda(s_1 \ldots s_t) = \sum s_1 \ldots s_{i-1} m_{s_i} s_{i+1} \ldots s_t$$

in M. Extending by A-linearity we obtain a biderivation. □

Proof. Let $P_\bullet \to B$ be a resolution of B over A (Remark 0.2). We will identify $L_{B/A}$ with $\Omega_{P_\bullet/A} \otimes B$. To construct the map we make some choices.

Choose an A-algebra map $\psi : P_0 \to P$ compatible with the given maps $P_0 \to B$ and $P \to B$.

Write $P_1 = A[S]$ for some set S. For $s \in S$ we may write

$$\psi(d_0(s) - d_1(s)) = \sum p_{s,t} f_t$$

for some $p_{s,t} \in P$. Think of $F = \bigoplus_{t \in T} P$ as a (P_1, P_1)-bimodule via the maps $(\psi \circ d_0, \psi \circ d_1)$. By Lemma 2.6 we obtain a unique A-biderivation $\lambda : P_1 \to F$ mapping s to the vector with coordinates $p_{s,t}$. By construction the composition

$$P_1 \longrightarrow F \longrightarrow P$$

sends $f \in P_1$ to $\psi(d_0(f) - d_1(f))$ because the map $f \mapsto \psi(d_0(f) - d_1(f))$ is an A-biderivation agreeing with the composition on generators.

For $g \in P_2$ we claim that $\lambda(d_0(g) - d_1(g) + d_2(g))$ is an element of Rel. Namely, by the last remark of the previous paragraph the image of $\lambda(d_0(g) - d_1(g) + d_2(g))$ in P is

$$\psi((d_0 - d_1)(d_0(g) - d_1(g) + d_2(g)))$$

which is zero by Simplicial, Section 0194).

The choice of ψ determines a map

$$\mathrm{d}\psi \otimes 1 : \Omega_{P_0/A} \otimes B \longrightarrow \Omega_{P/A} \otimes B$$

Composing λ with the map $F \to F \otimes B$ gives a usual A-derivation as the two P_1-module structures on $F \otimes B$ agree. Thus λ determines a map

$$\overline{\lambda} : \Omega_{P_1/A} \otimes B \longrightarrow F \otimes B$$

Finally, We obtain a B-linear map

$$q : \Omega_{P_2/A} \otimes B \longrightarrow Rel/TrivRel$$

by mapping $\mathrm{d}g$ to the class of $\lambda(d_0(g) - d_1(g) + d_2(g))$ in the quotient.

The diagram

$$\begin{array}{ccccccc}
\Omega_{P_3/A} \otimes B & \longrightarrow & \Omega_{P_2/A} \otimes B & \longrightarrow & \Omega_{P_1/A} \otimes B & \longrightarrow & \Omega_{P_0/A} \otimes B \\
\downarrow & & \downarrow{\scriptstyle q} & & \downarrow{\scriptstyle \overline{\lambda}} & & \downarrow{\scriptstyle \mathrm{d}\psi \otimes 1} \\
0 & \longrightarrow & Rel/TrivRel & \longrightarrow & F \otimes B & \longrightarrow & \Omega_{P/A} \otimes B
\end{array}$$

commutes (calculation omitted) and we obtain the map of the lemma. By Remark 2.3 and Lemma 2.2 we see that this map induces isomorphisms $H_1(L_{B/A}) \to H_1(L)$ and $H_0(L_{B/A}) \to H_0(L)$.

It remains to see that our map $L_{B/A} \to L$ induces an isomorphism $H_2(L_{B/A}) \to H_2(L)$. Choose a resolution of B over A with $P_0 = P = A[u_i]$ and then P_1 and P_2 as in Example 1.2. In Remark 2.5 we have constructed an exact sequence

$$\wedge^2_B(J_0/J_0^2) \to \mathrm{Tor}_2^{P_0}(B,B) \to H^{-2}(L_{B/A}) \to 0$$

where $P_0 = P$ and $J_0 = \mathrm{Ker}(P \to B) = I$. Calculating the Tor group using the short exact sequences $0 \to I \to P \to B \to 0$ and $0 \to Rel \to F \to I \to 0$ we find that $\mathrm{Tor}_2^P(B,B) = \mathrm{Ker}(Rel \otimes B \to F \otimes B)$. The image of the map $\wedge^2_B(I/I^2) \to \mathrm{Tor}_2^P(B,B)$ under this identification is exactly the image of $TrivRel \otimes B$. Thus we see that $H_2(L_{B/A}) \cong H_2(L)$.

Finally, we have to check that our map $L_{B/A} \to L$ actually induces this isomorphism. We will use the notation and results discussed in Example 1.2 and Remarks 2.5 and 2.4 without further mention. Pick an element ξ of $\mathrm{Tor}_2^{P_0}(B,B) = \mathrm{Ker}(I \otimes_P I \to I^2)$. Write $\xi = \sum h_{t',t} f_{t'} \otimes f_t$ for some $h_{t',t} \in P$. Tracing through the exact sequences above we find that ξ corresponds to the image in $Rel \otimes B$ of the element $r \in Rel \subset F = \bigoplus_{t \in T} P$ with tth coordinate $r_t = \sum_{t' \in T} h_{t',t} f_{t'}$. On the other hand, ξ corresponds to the element of $H_2(L_{B/A}) = H_2(\Omega)$ which is the image via $\mathrm{d} : H_2(\mathcal{J}/\mathcal{J}^2) \to H_2(\Omega)$ of the boundary of ξ under the 2-extension

$$0 \to \mathrm{Tor}_2^{\mathcal{O}}(\underline{B},\underline{B}) \to \mathcal{J} \otimes_{\mathcal{O}} \mathcal{J} \to \mathcal{J} \to \mathcal{J}/\mathcal{J}^2 \to 0$$

We compute the successive transgressions of our element. First we have

$$\xi = (d_0 - d_1)(-\sum s_0(h_{t',t} f_{t'}) \otimes x_t)$$

and next we have

$$\sum s_0(h_{t',t} f_{t'}) x_t = d_0(v_r) - d_1(v_r) + d_2(v_r)$$

by our choice of the variables v in Example 1.2. We may choose our map λ above such that $\lambda(u_i) = 0$ and $\lambda(x_t) = -e_t$ where $e_t \in F$ denotes the basis vector corresponding to $t \in T$. Hence the construction of our map q above sends $\mathrm{d}v_r$ to

$$\lambda(\sum s_0(h_{t',t} f_{t'}) x_t) = \sum\nolimits_t \left(\sum\nolimits_{t'} h_{t',t} f_{t'} \right) e_t$$

matching the image of ξ in $Rel \otimes B$ (the two minus signs we found above cancel out). This agreement finishes the proof. □

References

[1] S. Lichtenbaum and M. Schlessinger. The cotangent complex of a morphism. *Trans. Amer. Math. Soc.*, 128:41–70, 1967.